

SΔφ-65 — Slop as Externalized Restabilization Cost: AI Slop, Human Slop, Authority Pollution, and the Cost of Re-entry (v1.0)

Sofience · v1.0 · 2026-05-12 · DOI: 10.5281/zenodo.20127428

Root proposition: Slop is not defined by whether it is produced by AI or humans. Slop is produced when low execution or production TCC externalizes high verification, correction, interpretation, rebuttal, and restabilization TCC onto others. AI Slop tends to pollute search, recommendation, and content terrain. Human Slop tends to pollute judgment, authority, and common-sense terrain.

Abstract

SΔφ-65 defines Slop not by whether it is produced by AI or humans, but by whether low execution or production TCC externalizes high verification, correction, interpretation, rebuttal, and restabilization TCC onto others. The document distinguishes AI Slop and Human Slop, introduces authority-weighted rebuttal cost, formalizes World-Binding as a protection condition, and prevents over-classification of creative fiction by requiring fact-fiction confusion, externalized TCC to believers, and extracted benefit before strong Slop classification. SΔφ-65 connects SΔφ-56 v1.3 friction-adjusted TCC and SΔφ-64 v1.1 language-as-temporary-fixation into a practical audit protocol for platforms, media, AI systems, institutions, creators, and public reasoning.

Root Proposition

Slop is not defined by whether it is produced by AI or humans. Slop is produced when low execution or production TCC externalizes high verification, correction, interpretation, rebuttal, and restabilization TCC onto others. AI Slop tends to pollute search, recommendation, and content terrain. Human Slop tends to pollute judgment, authority, and common-sense terrain.

1. Definition of Slop

In SΔφ-65, Slop is defined as a transition-cost asymmetry rather than a creator category. The central error in many public uses of the term "AI Slop" is to treat AI generation as the decisive feature. SΔφ instead asks: who pays the cost after low-cost production or speech enters the world?

Slop appears when an output is cheap to produce, easy to distribute, weakly bound to the world, difficult to rebut or clean up, and beneficial to the producer while shifting verification, interpretation, correction, or restabilization costs to others.

Minimal definition:

Slop = Low execution or production TCC + weak World-Binding + externalized restabilization TCC.

Strong harmful definition:

Harmful Slop = fact-fiction confusion + significant TCC to those who believe it + benefit to a producer, group, institution, or platform.

2. AI Slop

AI Slop is not merely AI-generated content. AI-generated art, simulation, education, animation, fiction, visualization, or assistance may be legitimate when its status, limits, and revision path are clear.

AI Slop occurs when low production TCC, automation, repetition, weak World-Binding, and low accountability shift cleanup, selection, verification, interpretation, and platform restabilization costs to others.

Formula:

$AI_Slop_Risk = Low_Production_TCC \times Volume \times (1 - World_Binding) \times Low_Accountability \times Cleanup_TCC.$

Typical contaminated terrain:

- search results
- recommendation systems
- children-facing content surfaces
- data sets
- platform trust
- creator reward systems
- user attention and verification capacity

AI Slop is therefore primarily a volume-and-terrain pollution problem: production is cheap, but filtering and restabilization are expensive.

3. Human Slop

Human Slop is produced when low speech TCC is amplified by authority, distribution, institutional backing, or public trust while weak World-Binding and high rebuttal cost are shifted to others.

Human Slop is not simply a human being making a mistake. It occurs when authority-weighted speech makes the cost of refusal, rebuttal, correction, and social restabilization larger than the speaker can or will absorb.

Formula:

$Human_Slop_Risk = Low_Speech_TCC \times Authority_Weight \times Domain_Distance \times (1 - World_Binding) \times Rebuttal_TCC \times Restabilization_TCC.$

Human Slop tends to contaminate:

- public judgment
- institutional default assumptions
- common sense
- political or religious belief fields
- media frames
- expert trust
- organizational decision-making

AI Slop pollutes content terrain. Human Slop pollutes judgment terrain.

4. Authority and Rebuttal TCC

Authority is not the same as truth. In $S\Delta\phi$, authority is the power of a person, institution, narrative, symbol, platform, or record to bind another agent's judgment path and increase the TCC of rebuttal, refusal, correction, or exit.

Authority may come from:

- formal office
- academic achievement
- professional reputation
- institutional endorsement
- legacy media distribution
- public familiarity
- symbolic status
- religious or ideological belonging
- platform ranking or visibility
- archive and citation persistence

A fiction or falsehood may have high social authority if many people treat it as fact and if rebutting it imposes high TCC.

$Authority\ Weight = Recognition + Institutional\ Backing + Distribution\ Power + Default-Setting\ Power + Rebuttal\ TCC + Record\ Persistence + Sanction/Exclusion\ Power.$

Authority becomes risky when it compensates for weak World-Binding. A claim with weak evidence but strong authority may travel farther than a well-bound correction.

5. World-Binding Override Rule

Slop is not novelty, strangeness, controversy, or low initial acceptance. A claim must not be classified as Slop merely because it is new, unpopular, disruptive, or outside current consensus.

World-Binding protects a claim from Slop classification.

World-Binding Override Rule:

Even if a statement has low execution TCC, high dissemination, high authority weight, or high rebuttal cost, it should not be classified as Slop when it demonstrates, or is strongly expected to demonstrate, high World-Binding through reproducible observation, predictive power, experimental robustness, explanatory integration, or durable correction capacity.

Provisional strong World-Binding may be inferred only when the claim:

1. generates testable predictions,
2. exposes itself to possible falsification,
3. preserves revision paths,
4. does not rely solely on authority,
5. reduces explanatory or operational cost over time,
6. allows independent observers to inspect or reproduce its binding conditions.

Newtonian gravity, Einsteinian relativity, germ theory, quantum theory, and other historically disruptive frameworks are not Slop merely because they were once strange or contested. Their authority was not the source of their validity. Their World-Binding later generated authority.

6. Creative Fiction Protection Rule

SΔφ-65 must not punish fiction, art, satire, myth, speculative design, simulation, game worlds, or creative imagination merely because they are not strongly World-Bound.

Creative fiction is not Slop when its fictional status remains available to the audience and when it does not intentionally collapse fact and fiction in order to extract benefit while shifting large TCC to believers.

Creative Fiction Protection Rule:

Creative fiction, art, myth, satire, simulation, and speculative works should not be classified as Slop merely because they are not world-bound. They become Slop only when they are presented or routed in a way that intentionally collapses the distinction between fiction and fact, externalizes significant TCC onto those who believe them, and generates benefit for a producer, institution, group, or platform.

Strong harmful Slop requires three conditions:

1. It intentionally induces confusion between fact and fiction.
2. Those who judge it as fact receive significant TCC.
3. The confusion generates benefit for a specific individual, group, institution, platform, or coalition.

Fiction opens worlds. Slop disguises fiction as fact and externalizes the cost of that disguise.

7. Restabilization TCC Explosion

The core danger of Slop is not the initial error. The danger is restabilization TCC explosion.

Restabilization_TCC > Executor_Capacity.

A speaker, creator, bot, platform, or institution may produce a low-cost output, but once that output enters search, media, memory, policy, identity, or institutional pathways, the cost of correcting and restabilizing the field may exceed the executor's capacity to repair it.

AI Slop example:

- one automated channel produces thousands of weakly bound videos;
- platform and users must filter, report, downrank, correct, and restore trust.

Human Slop example:

- an authority figure states a weakly bound claim in a broadcast;
- specialists, audiences, institutions, and affected groups must absorb rebuttal and restabilization costs.

Slop becomes dangerous when the entity that benefits from low execution cost cannot or does not carry the high downstream restabilization cost.

8. Slop Classification Levels

SΔφ-65 distinguishes weak, harmful, and sanction-grade Slop.

Weak Slop:

Low-cost low-quality output that increases filtering or interpretation burden, but may not intentionally deceive or produce large harm.

Harmful Slop:

Fact-fiction confusion + externalized TCC + extracted benefit.

Sanction-grade Slop:

Harmful Slop with one or more escalation factors: large-scale dissemination, vulnerable audience targeting, institutional authority, repetition after correction, significant financial/political gain, identity-based harm, health/safety impact, or irreversible restabilization cost.

This tiering prevents overclassification. Not all weak outputs are sanction-grade. Not all fiction is Slop. Not all novel theories are Slop. But repeated fact-fiction collapse with externalized cost and extracted benefit must be treated as a serious audit object.

9. AI Slop and Human Slop Comparison

AI Slop and Human Slop share the same deep structure: low execution TCC externalizes high restabilization TCC. They differ in primary amplification route.

AI Slop:

- main amplifier: automation and volume
- main contamination: search, recommendation, content terrain
- main externalized costs: filtering, verification, cleanup, data set hygiene, attention depletion

Human Slop:

- main amplifier: authority and trust
- main contamination: judgment, default assumptions, common sense, institutional memory
- main externalized costs: rebuttal, correction, public education, trust repair, social restabilization

Compressed distinction:

AI Slop is quantity-driven terrain pollution.

Human Slop is authority-driven judgment pollution.

10. Operational Kernel Use

When SΔφ Operational Kernel evaluates a suspected Slop case, it should not begin by asking whether the content was generated by AI. It should ask:

1. Is this a fact claim, fiction, satire, speculation, simulation, or mixed presentation?
2. Is the fact/fiction boundary clear to the expected audience?
3. What is the World-Binding strength?
4. If World-Binding is weak, is authority, automation, or distribution compensating for that weakness?
5. Who receives verification, rebuttal, interpretation, correction, or restabilization TCC?
6. Who benefits from the confusion, volume, authority, or dissemination?
7. Is there a revision, correction, or re-entry path?
8. Does restabilization TCC exceed executor capacity?

Default judgment:

If World-Binding is strong, do not classify as Slop. If fictional status is clear, do not classify as Slop. If fact-fiction confusion is induced, believers receive large TCC, and the producer benefits, classify as Harmful Slop. If authority or volume makes correction effectively impossible, escalate toward Sanction-grade Slop.

11. Do-Not-Use Conditions

Do not use SΔφ-65 to dismiss artistic fiction, mythology, satire, speculative design, game worlds, or experimental media merely because they are not factual.

Do not use SΔφ-65 to attack novelty or unconventional theory merely because it lacks early consensus.
Do not classify a claim as Slop solely because it is AI-generated.
Do not classify a claim as Human Slop solely because a non-expert speaks about a field outside their expertise. Classification requires low World-Binding or weak evidence, authority or dissemination compensation, high rebuttal/restabilization TCC, and externalized cost.
Do not use Slop classification as a censorship shortcut. Use it as a cost-attribution and re-entry audit.

12. Minimal Output Template

SΔφ Slop Audit Minimal Output:

- Claim/content type: fact / fiction / satire / speculation / mixed / unclear
- Production or speech TCC: low / medium / high
- World-Binding: weak / moderate / strong / unknown
- Authority or volume amplifier: none / low / medium / high
- Fact-fiction confusion: absent / ambiguous / induced
- TCC to believers or receivers: low / medium / high
- Benefit to producer or group: none / indirect / direct
- Rebuttal/restabilization TCC: low / medium / high / exceeds executor capacity
- Classification: not Slop / weak Slop / harmful Slop / sanction-grade Slop / UMR pending
- Revision path: available / weak / blocked

AI Slop vs Human Slop Summary Table

Axis	AI Slop	Human Slop
Primary amplifier	Automation and volume	Authority and trust
Execution cost	Low production TCC	Low speech TCC
Polluted terrain	Search, recommendation, content, data sets	Judgment, common sense, institutional memory
Main rebuttal burden	Filtering and cleanup	Authority-weighted rebuttal and restabilization
Core risk	Quantity-driven terrain pollution	Authority-driven judgment pollution
Strong Slop condition	Fact-fiction confusion + externalized TCC + benefit	Fact-fiction confusion + externalized TCC + benefit